

Academic Qualifications			
Year	Degree/Certificate	Institute	CPI/%
2024-Present	B.Tech	Indian Institute of Technology Kanpur	8.7/10
2024	JKBOSE(XII)	Burn Hall School, Sonwar, Srinagar	93.6%
2022	JKBOSE(X)	Burn Hall School, Sonwar, Srinagar	96.8%

Scholastic Achievements
Placed in the Center Top 10% for the NSEC, India’s foremost national Chemistry Olympiad examination.

Work Experience	
Founder’s Office Intern - Product and Growth Cherrie Date   (Dec’25 - Feb’26)	
Objective	Spearheaded the zero-to-one market entry for a safety-centric dating platform to redefine the local ecosystem.
Approach	- Formulated a Go-to-Market (GTM) strategy leveraging experiential marketing and offline mixer events. - Drove Women First user acquisition through safety propositions of Time Compensation & Ironclad Safety.
Impact	Established an initial market footprint by converting offline engagement into a verified early-adopter pipeline.

Research Experience	
Multimodal Machine Learning for Early Detection of Major Depressive Disorder (MDD) Using Neuroimaging and Physiological Biomarkers Prof. Hamim Zafar CoSmic Lab SURGE ML - Intern IIT Kanpur  (May’26 - Present)	
Objective	- To develop an end-to-end multimodal learning pipeline for early detection of Major Depressive Disorder. - To integrate resting-state fMRI, structural MRI, and clinical biomarkers for HAMD score prediction.
Approach	- Built a multimodal learning pipeline using rs-fMRI, structural MRI, and biomarkers for MDD detection. - Engineered a REST-meta-MDD cohort (2,380 subjects and 25 sites) through phenotype harmonization. - Developed a parallel Nilearn engine for large-scale ROI and Nifti feature extraction across the full cohort. - Extracted 7,250+ features per subject, including connectivity features and ROI-wise neuroimaging metrics. - Designed a leakage-free late-fusion architecture integrating functional, structural, and clinical parameters. - Validated using leave-one-site-out cross-validation and ComBat harmonization for MDD prediction.
Result	- Curated a harmonized cohort of 1,707 subjects from the original 2,380-subject multi-site release. - Consolidated 7,250 neuroimaging features per subject into standardized analysis-ready master dataset. - Enabled robust MDD classification and HAMD regression through multimodal feature extraction.

Key Projects	
Deepfake Detection EE-656 (Course Project) Prof. Nischal Kumar Verma IIT Kanpur  (May’26 - Present)	
Objective	Benchmark deepfake detection systems to assess accuracy, efficiency, and robustness under media degradations.
Approach	- Built a deepfake detection pipeline benchmarking MesoNet-4 and Xception using frames from Celeb-DF v2. - Trained both models for 30 epochs on 6,392 face images generated from curated Celeb-DF v2 subjects. - Evaluated AUC-ROC, inference latency, FLOPs, and robustness under Gaussian noise & JPEG compression.
Results	Achieved AUC-ROC of 0.90/0.92 and identified MesoNet for edge deployment and Xception for robustness.

Brain spy: Deciphering Alzheimer’s Brain and Cognitive Science Club IIT Kanpur  (May’25 - Sep’25)	
Objective	- Develop an end-to-end deep learning framework for early-stage Alzheimer’s Disease detection. - Leverage 3D structural MRI for automated disease classification. - Incorporate attention-based explainability to improve model interpretability.
Approach	- Engineered an MRI preprocessing pipeline on ADNI-1 (1,891 scans, 650 subjects) dataset for Alzheimer’s. - Leveraged Neural Pre-Processing (NPP) for joint skull stripping, bias correction, and MRI normalization. - Reduced MRI preprocessing time by 9× relative to the SynthStrip baseline. - Designed a modified AXIAL explainable CNN on a frozen ResNet152 backbone with Attention-XAI Fusion. - Produced 2048-dimensional embeddings with localized diagnostic saliency attention-guided feature fusion. - Developed ThreeHeadADNet, a multi-view dual-CNN integrating axial, coronal, and sagittal slices. - Benchmarked against 10+ architectures, including ResNet152, VGG19, ADNet3D, DHO-CapsNet and AXIAL.
Result	- Achieved 97% test accuracy (Macro F1 = 0.965) with Modified AXIAL and 96.67% accuracy (F1 = 0.9667). - Enforced strict subject-wise validation, retaining 85%/84% accuracy while eliminating train-test leakage. - Identified and corrected a fatal data-leakage issue in a reported 99% accuracy pipeline research paper.

Objective	• To develop a Deep Neural Network Compression framework for compressing convolutional architectures. • To achieve efficient edge deployment via model-size reduction while maintaining near-baseline performance.
Approach	• Trained a VGG-11 CNN on CIFAR-10 baseline and developed a modular three-stage compression pipeline. • Reduced model footprint through magnitude-based iterative pruning while avoiding retraining from scratch. • Achieved 90% sparsity and compressed weights using K-Means quantization with cluster weight-sharing. • Applied Huffman coding to maximize bit-level compression of sparse weight matrices. • Performed mask-preserving fine-tuning to recover post-pruning accuracy and maintain model performance. • Benchmarked top-1 accuracy, sparsity ratio, and compressed model size across all pipeline stages for traceability.
Results	• Achieved 23.8× model compression (37.22 MB → 1.57 MB) through pruning, quantization, and coding. • Limited performance degradation to only 0.68% accuracy drop, retaining 89.99% top-1 accuracy. • Demonstrated production-ready edge deployment through memory reduction with near-baseline accuracy.

Objective	• Architecting a PM2.5 forecasting pipeline and LLM-driven recommendations across 31 NCR stations. • Building a city-scale environmental intelligence system combining graph forecasting & LLM's. • Developing a recommendation engine leveraging PM2.5 prediction and natural-language reasoning. • Engineering an end-to-end decision-support platform transforming forecasts into health-aware guidance.
Approach	• Built a graph-based spatio-temporal forecaster with probabilistic confidence estimation for PM2.5 prediction. • Designed a multi-agent backend integrating route-planning & notification agents for real-time support.
Results	• Delivered an adaptive LLM-powered alerting framework converting probabilistic forecasts into healthy routes.

Objective	• Engineering a cluster-aware PM2.5 forecasting pipeline across 31 Delhi NCR stations time-series data.
Approach	• Built a data pipeline with interpolation, outlier clipping, resampling & feature engineering for NAQI data. • Designed a spatial-clustering framework using PM2.5–meteorology correlations, PCA, and K-Means. • Trained & tested 93 station models using Random Forest, XGBoost, & Decision Trees with validation.
Results	• Achieved $R^2 = 0.75$ (peak) and $R^2 = 0.40$ (mean) across 31 Delhi NCR stations using Random Forest.

Objective	• Develop a hybrid ARIMA–LSTM framework for equity-price forecasting, combining linear trend extraction with nonlinear residual learning. • Model both long-term market trends and short-term temporal dynamics.
Approach	• Built an end-to-end equity data pipeline using yfinance, handling missing values, outliers, and stationarity. • Applied ADF testing with ACF/PACF tuning to build an ARIMA trend model in statsmodels. • Trained a gated LSTM in TensorFlow/Keras on ARIMA residuals to learn nonlinear dynamics. • Integrated ARIMA and LSTM with strict time-aware train/test validation. • Benchmarked performance using RMSE, MAE, MAPE, and directional accuracy.
Results	• Achieved 4.31% MAPE (Nifty) and 6.69% MAPE (TCS), outperforming standalone ARIMA and LSTM. • Validated robustness through residual diagnostics and trend analysis.

Objective	• Engineer an end-to-end ML pipeline for symptom-based disease diagnosis with explainable predictions.
Approach	• Built a reproducible ETL pipeline integrating 8 medical datasets into a clean 4,920 × 132 feature matrix. • Engineered a bag-of-symptoms representation with LabelEncoder and stratified 80/20 validation. • Trained a 300-tree Random Forest using bagging, feature subsampling, and balanced class weighting. • Deployed a stateless Flask REST API serving probabilistic predictions and top-5 differential diagnoses.
Results	• Delivered explainable diagnosis using confidence scores, and symptom-severity ranking across 41 diseases. • Achieved Macro-F1 = 1.00 on a 984-sample test set with sub-second inference and 100% KB coverage.

Problem	• Assess environmental risks across an 810,000 sq ft, 550-bed medical school, impacting 8,000+ stakeholders.
Approach	• Developed an integrated environmental strategy using risk prioritisation and phased execution planning.
Results	• Delivered a 47-page implementation roadmap, projected to significantly mitigate environmental risks.

- Engineered a **dual-motor continuous-track stair-climbing robot** in a **6-member team** for payload transport over staircases.
- Designed an **Arduino-controlled bevel-gear lifting mechanism** for adaptive stair negotiation and secure load handling.
- Manufactured & assembled the prototype using **3-D printing, drilling, milling, & soldering**, integrating Mech & Elec subsystems.

- Engineered a **scaled working model** of the London Bridge through **metal fabrication**, replicating its structural mechanism.
- Fabricated **mild-steel** and **galvanized-iron** members using **sheet-metal forming, shearing, and drilling** for accurate assembly.
- Assembled the structure using **brazing, welding, and riveting** to achieve a rigid and durable final prototype.

Technical Skills

- Programming Languages:** C, C++, HTML, CSS, L^AT_EX, Python, SQL
- Libraries:** OpenCV, NumPy, Pandas, Matplotlib, TensorFlow, Scikit-Learn, PyTorch, Nibabel, ipywidgets, Seaborn, Nilearn
- Softwares:** CAD Fusion 360, AutoCAD, Simulink, MATLAB, Ansys, Mathematica, Arduino, Docker

Fundamentals of Computing	Probability & Statistics	Linear Algebra	(*) Introduction to Electronics
Oscillations and Waves	(*) Environ. & Sustainability	Fluid Mechanics	AI, ML, DL & Applications
Partial Differential Equations	(*) Mechanics of Solids	(*) Soil Mechanics (A*)	(*) Multivariable Calculus
Classical Dynamics	Ordinary Differential Equations	Engineering Graphics	Civil Engg. Materials
(*) Adv. English Communication	(*) Organic Chemistry Fundamentals and Applications	(*) Environ. Quality & Processes	Physical Chemistry

Positions of Responsibility

Leadership	- Elected by a 600+ electorate to represent Hall 3 at all institutional levels, including CoSHA and HMC. - Led Hall 3 to victory in the inter-hall general championship, coordinating over 1500+ individuals. - Comprehensively drafted a 15+ lakh budget, ensuring optimal utilization of DCF, HDF and Ext. Funds. Coordinated daily hall operations with the Hall Manager, supervising 7 sanitation and 5 horticulture workers.
Initiatives	Mobilized ₹ 20 Lakh+ in funding to renovate 9 hall washrooms, improving resident hygiene standards. - Directed a ₹ 1 Lakh+ Reading Room revamp initiative, creating an improved academic workspace for residents.
Impact	Led Hall 3 to GC 2025–26 victory, securing titles in Galaxy, Inferno, and Freshers' Inferno.

Leadership	- Coordinating within a 4-tier managing a forecasted budget of INR 2 Cr+ to organize Antaragni'26 - Collaborating with the core team from 29 administrative portfolios to organize the cultural festival - Drove pre-fest planning, social outreach, and external collaborations to support Antaragni'26 execution.
Work	- Onboarded Sponsorship and negotiated collaboration deals worth INR 7 Lakhs + for Antaragni,26 - Conducting Roadtrips in 7+ Metrocities of India Partnering with 50+ bands pulling 50k+ footfall.
Impact	Collaborated with State ministries and foreign embassies for international promotion of Antaragni'26.

- Spearheaded a **750-member** hall contingent in **Takneek**, IITK's flagship inter-hall technological championship, orchestrating end-to-end strategy to steer the contingent to a **2nd-place overall finish**.
- Strategically marshalled participation across **11 problem statements** spanning diverse technical domains, overseeing team formation, mentorship and execution to maximize cross-event scoring.
- Administered **end-to-end financial budgeting and resource allocation** for the contingent, ensuring optimal capital deployment across all 11 competing problem statements.

- Mentored and guided **5 first-year students** through a full academic year, easing their transition into IITK's academic rigor and campus culture for a smoother freshman experience.
- Counselled freshers throughout their **orientation period**, delivering structured academic and social guidance to ensure a confident **school-to-college** transition.
- Organized **orientation events** and managed **hospitality logistics**, coordinating end-to-end arrangements to deliver a welcoming, well-run induction for incoming students.

Social Impact | Community Education Volunteer | Kashmir Foundation for Organisation R&D | (Jun'25 - Jul'25)

Objective	● Improve foundational literacy & numeracy among tribal children through strategic community-led education.
Approach	● Designed a structured curriculum across 8 classroom sessions for 30 Gujjar/Bakarwal children aged 6–12.
Impact	● Increased attendance from 8 to 30 students through targeted community engagement and awareness efforts. ● Developed curriculum-scaling strategies and future community-engagement strategies for sustained impact .

Lab Experience

ESC201 | Introduction to Electronics | IIT Kanpur (Jan'26 - Apr'26)

Mentor	Prof. A.K.R. Sivakumar & Prof. Rik Dey, Dept. of EE, IIT Kanpur (Grade: A)
Work	● Characterized the time- and frequency-domain response of RC and RL circuits, extracting resonance frequency and Q-factor of an RLC network through systematic bench measurement and analysis. ● Realized a half-adder and the CARRY function of a Full Adder using ICs 7486 (EX-OR), 7400 (NAND) and 7402 (NOR) , validating Boolean logic synthesis against theoretical truth tables.

CE252 | Soil Mechanics | IIT Kanpur (Jan'26 - Apr'26)

Mentor	Prof. Bipin Gupta, Dept. of Civil Engineering, IIT Kanpur (Grade: A*)
Work	● Conducted geotechnical experiments to determine core soil properties including plasticity, compaction, permeability and shear strength across varied soil samples under controlled lab conditions. ● Analyzed consolidation behavior and drainage characteristics through oedometer testing , empirically validating Terzaghi's consolidation theory against measured settlement data. ● Interpreted direct-shear and triaxial test results using Mohr's Circle to construct failure envelopes and characterize the shear-strength behavior of cohesive and granular soils.

CE272 | Fluid Mechanics for Civil Engineers | IIT Kanpur (Aug'25 - Dec'25)

Mentor	Prof. Chunendra Sahu & Prof. Shivam Tripathi, Dept. of Civil Engineering, IIT Kanpur
Work	● Conducted experiments on flow-measurement devices including venturimeters and orificemeters to validate Bernoulli's equation against measured discharge coefficients. ● Analyzed laminar and turbulent flow regimes through Reynolds-number estimation and pressure-drop evaluation across varied pipe networks to characterize transitional flow behavior. ● Verified core fluid principles — continuity, momentum conservation and Navier-Stokes assumptions — through controlled laboratory measurement and comparison against analytical predictions.

Extracurricular Activities

Debating	● Earned a Verbal Mention at IIMUN Srinagar '19 (Lok Sabha) for analytical reasoning and public speaking.
Leadership	● Managed logistics and a ₹ 1 Lakh budget for Maurya Bhoj , a large-scale residential dining & felicitation event. ● Led extracurricular and sports participation as House Captain of Shakespeare House across Grades 4–10. ● Led 11 technical problem statements simultaneously, as Hall Captain in the Intra-IIT Tech. Competition.
Mentoring	● Guided freshers through academic and social transition as a Student Guide with a 22-member core team . ● Mentoring the AirSense AI project project under SOCE , developing an AI-driven air-quality decision system.
Sports	● Won consecutively a Gold Medal in Carrom at Inferno '26 and a Silver Medal in Carrom at Inferno '25 . ● Won Gold Medal in Weightlifting (59–63 kg) and Silver Medal in Football at Freshers' Inferno '24 .
Cultural	● Awarded a Silver Medal for performance in the Galaxy '26 Parade for exemplary coordination and execution.
Social	● Ex-Officio Member of the Anti-Ragging Committee , Hall III, ensuring resident safety and campus well-being
Awards	● Won the Best Research Project Award from the Science & Technology Council IITK for project BrainSpy .